

Basics of Reinforcement Learning

1 Multi-Armed Bandits and Value Estimation

Let A_t denote the action selected at time step t , and $R_t \in \mathbb{R}$ be the corresponding reward. The expected reward for an action a is defined as:

$$Q(a) = \mathbb{E}[R_t \mid A_t = a] \quad (1)$$

The objective is to maximize the expected cumulative reward over time T :

$$\max \sum_{t=1}^T \mathbb{E}[R_t] \quad (2)$$

The sample-average estimate of the action value $\hat{Q}_t(a)$ is given by:

$$\hat{Q}_t(a) = \frac{\sum_{i=1}^{t-1} R_i \cdot \mathbb{I}(A_i = a)}{\sum_{i=1}^{t-1} \mathbb{I}(A_i = a)} \quad (3)$$

where $\mathbb{I}(\cdot)$ is the indicator function. Under a greedy policy, the agent selects:

$$A_t = \arg \max_a \hat{Q}_{t-1}(a) \quad (4)$$

2 Markov Decision Process (MDP)

A sequence $\{S_t, A_t, R_t\}_t$ forms a Markov Decision Process if and only if the transition dynamic satisfies the Markov property:

$$\Pr(S_{t+1}, R_t \mid A_t, S_t) = \Pr(S_{t+1}, R_t \mid A_t, S_t, R_{t-1}, A_{t-1}, S_{t-1}, \dots) \quad (5)$$

For stationary transition dynamics:

$$\Pr(S_{t+1}, R_t \mid A_t = a, S_t = s) = \Pr(S_t, R_{t-1} \mid A_{t-1} = a, S_{t-1} = s) \quad (6)$$

The agent implements a policy mapping history $\bar{S}_t$ to a probability distribution over actions:

$$\pi_t(a \mid s) = \Pr(A_t = a \mid \bar{S}_t = s), \quad \text{where } \bar{S}_t = (S_t, R_{t-1}, A_{t-1}, \dots, R_0, A_0, S_0) \quad (7)$$

For a **Markov policy**, π_t depends on $\bar{S}_t$ only through the current state S_t .

3 Return and Value Functions

The goal of the agent is to maximize the expected return G_t .

- **Average Return:**

$$G_t = \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{i=t}^{T-1} R_i \quad (8)$$

- **Discounted Return:**

$$G_t = \sum_{i=0}^{\infty} \gamma^i R_{t+i} = R_t + \gamma R_{t+1} + \gamma^2 R_{t+2} + \dots \quad (9)$$

where $\gamma \in [0, 1)$ is the discount factor.

The **State-Value Function** $V^\pi(s)$ under policy π is:

$$V^\pi(s) = \mathbb{E}^\pi [G_t \mid S_t = s] = \mathbb{E}^\pi \left[\sum_{i=0}^{\infty} \gamma^i R_{t+i} \mid S_t = s \right] \quad (10)$$

The **Action-Value Function** $Q^\pi(s, a)$ is:

$$Q^\pi(s, a) = \mathbb{E}^\pi [G_t \mid A_t = a, S_t = s] = \mathbb{E}^\pi \left[\sum_{i=0}^{\infty} \gamma^i R_{t+i} \mid A_t = a, S_t = s \right] \quad (11)$$

The relation between $V^\pi(s)$ and $Q^\pi(s, a)$ is:

$$V^\pi(s) = \sum_{a \in \mathcal{A}} \pi(a \mid s) Q^\pi(s, a) \quad (12)$$

4 Bellman Equations

The Bellman Equation for $V^\pi(s)$ decomposes value recursively:

$$V^\pi(s) = \mathbb{E}^\pi [R_t + \gamma V^\pi(S_{t+1}) \mid S_t = s] \quad (13)$$

$$= \mathbb{E}^\pi [R_t + \gamma(R_{t+1} + \gamma R_{t+2} + \dots) \mid S_t = s] \quad (14)$$

$$= \mathbb{E}^\pi [R_t \mid S_t = s] + \gamma \mathbb{E}^\pi [G_{t+1} \mid S_t = s] \quad (15)$$

$$= \mathbb{E}^\pi [R_t \mid S_t = s] + \gamma \mathbb{E}^\pi [\mathbb{E}^\pi [G_{t+1} \mid S_{t+1}] \mid S_t = s] \quad (16)$$

$$= \mathbb{E}^\pi [R_t \mid S_t = s] + \gamma \mathbb{E}^\pi [V^\pi(S_{t+1}) \mid S_t = s] \quad (17)$$

The Bellman Optimality Equation is given by:

$$V^{\pi_{\text{opt}}}(s) = \max_a \mathbb{E} [R_t + \gamma V^{\pi_{\text{opt}}}(S_{t+1}) \mid A_t = a, S_t = s] \quad (18)$$

5 Dynamic Programming Iterations

- **Policy Iteration:** $\pi_0 \rightarrow V^{\pi_0} \rightarrow \pi_1 \rightarrow V^{\pi_1} \rightarrow \dots \rightarrow \pi_{\text{opt}}$
- **Value Iteration:** $V_0 \rightarrow V_1 \rightarrow V_2 \rightarrow \dots \rightarrow V^{\pi_{\text{opt}}} \rightarrow \pi_{\text{opt}}$